Network-based diversification of stock portfolios with PortNet

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Abstract

Network science provides a sophisticated lens for examining asset interdependencies in financial markets, offering robust alternatives to traditional diversification methods that rely solely on linear correlation. This study replicates and extends the *network-based portfolio diversification methodology* proposed by Kitanovski et al. [1], which seeks to maintain an optimal balance between portfolio returns and risk amidst market volatility. We adapt their approach to constructing highly diversified portfolios of *S&P 500 stocks*, specifically leveraging a two-year in-sample period to build a stable network structure. The core method involves creating a *Minimal Spanning Tree (MST)* from asset correlations and stabilizing the resulting community structure through *monthly co-occurrence analysis*. We segment the market using the *Louvain community detection algorithm* and, for extension, the *Girvan-Newman algorithm*. Assets are selected based on topological metrics, including *Degree and Closeness Centrality*, and statistical metrics like *Principal Component Analysis (PCA)* loadings. The resulting portfolios are rigorously evaluated over a one-year out-of-sample period. The objective is to validate the efficacy of the core methodology and determine whether alternative community detection strategies, such as Girvan-Newman, can yield superior or more robust portfolio performance. **TODO:** *Initial results confirm that the centrality-based asset selection consistently yields lower volatility than the market baseline, and the Girvan-Newman-derived portfolios demonstrate comparable resilience under unseen data.* This work establishes a solid benchmark for adaptive, network-based portfolio construction, emphasizing the necessity of stabilized community identification in volatile financial systems.

1 Introduction

Portfolio diversification, rooted in the seminal work of Markowitz [3], is the cornerstone of risk mitigation in investment theory. Historically, this strategy relied on analyzing *linear relationships* between assets using the Pearson correlation coefficient. However, the intricacies of modern financial markets and periods of high volatility necessitate methodologies that can model the *non-linear dynamics* and complex interdependencies that traditional models fail to capture [1].

The *network science* approach, which models financial assets as nodes and their relationships as weighted edges, provides a robust alternative. This paradigm allows for the extraction of latent market structures, such as hierarchical organization [2] and distinct asset communities. Our work builds directly upon the *network-based portfolio diversification methodology* developed by Kitanovski et al. [1], which uses a refined community detection process to inform asset selection based on network topology and statistical properties.

In this report, we present *PortNet*, an implementation and empirical extension of this methodology, focusing on constructing highly diversified portfolios of *S&P 500 stocks*. **(TODO)**

Our empirical evaluation, conducted over a one-year out-of-sample period, demonstrates **(TODO)**. This paper is organized as follows: Section 2 covers related work, Section 3 details the PortNet methodology, Section 4 presents our results, and Section 5 concludes with a discussion of the findings and future work.

2 Related Work

TODO

3 Methods

TODO

4 Results

TODO

5 Discussion & Conclusion

TODO

References

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